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Permit B Holder

Sotirios Karapiperis

Education

Department of Computer Science, ETH Zurich — Zurich, Switzerland
MSc. in Computer Science — 2024 - 2026

- Advisor: Prof. Francis Engelmann
- Major: Visual & Interactive Computing.
- Thesis: Joint Structural and Geometric Generation of Polygonal Floorplans via Flow Matching
- Current GPA: 5.56/6.00

Department of Electical & Computer Engineering, NTU Athens — Athens, Greece
MEng. in Electrical & Computer Engineering — 2018 - 2023

- Advisor: Prof. Petros Maragos
- Diploma Thesis: 3D Human Shape and Pose Estimation from RGB Input.
- GPA: 9.4/10.0, Rank: 10/299

Publications

1. **Sotirios Karapiperis**, Nikolaos Ellinas, Alexandra Vioni, Junkwang Oh, Gunu Jho, Inchul Hwang, Spyros Raptis. Investigating disentanglement in a phoneme-level speech codec for prosody modeling. *IEEE Spoken Language Technology Workshop*, 2024.

Research Experience

Samsung | Athens, Greece — 03.2023 - 09.2024

- Research & Development of generative models for neural speech synthesis for the next generation of Samsung's text-to-speech product leading to a publication to IEEE SLT 2024.

DeepLab | Athens, Greece — 06.2022 - 08.2022

- Research & Development of generative models for the task of Few Shot Visual Relationship Detection.

Selected Projects

Estimating Articulated Objects | Semester Project @ ETH Zurich

- Developed a novel method for 3D point cloud segmentation, enabling the extraction of hierarchical object parts and estimation of joint parameters.

Mass-Spring 2DGS | Course Project @ ETH Zurich

- Enhanced the SpringGauss framework for reconstructing dynamic objects from multiple images by integrating 2D Gaussian Splatting (2DGS) to improve geometric accuracy, alongside developing a novel approach for handling object rotations. Project Page

2DGS/GoF-Avatar | Course Project @ ETH Zurich

- Building upon 3DGS-Avatar, proposed a more geometrically accurate approach using 2D Gaussian Splatting and Gaussian Opacity Fields.

Awards and Honors

- **Academic Scholarship** Mavroulias foundation scholarship for excellence in Pan-hellenic exams. 2019
- **3rd Place**, Hackathon competition held by SICK, IEEE NTUA SB. 2018
- **Academic Scholarship**, Union of Greek Shipowners. 2024
- **Academic Scholarship**, Onassis Foundation. 2024

Skills

Languages: Greek (Native), English (Fluent), German (Elementary)
Programming: Python (Strong), C/C++ (Strong), Matlab
Frameworks: Pytorch (Strong), Keras, Sklearn, Numpy, Matplotlib (Strong)

TA Experience

Lab Assistant Course: "Introduction to Computer Programming", NTUA 2020-2021
Teaching Assistant Course: "Visual Computing", ETHZ 2026
Teaching Assistant Course: "Computer Vision", ETHZ 2026
Teaching Assistant Course: "Mixed Reality", ETHZ 2026

Volunteering

Volunteer at ICASSP 2023 Rhodes,Greece 2023

Hobbies

Sailing, Running, Automation Projects

Arxiv Preprints

1. **Sotirios Karapiperis**, Markos Diomataris, Vassilis Pitsikalis. RelVAE: Generative Pretraining for few-shot Visual Relationship Detection. http://arxiv.org/abs/2311.16261 2023